

Derm-T2IM: Harnessing Synthetic Skin Lesion Data via Stable Diffusion Models for Enhanced Skin Disease Classification using ViT and CNN

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Abstract—This study explores the utilization of Dermatoscopic synthetic data generated through stable diffusion models as a strategy for enhancing the robustness of machine learning model training. Synthetic data generation plays a pivotal role in mitigating challenges associated with limited labeled datasets, thereby facilitating more effective model training. In this context, we aim to incorporate enhanced data transformation techniques by extending the recent success of few-shot learning and a small amount of data representation in text-to-image latent diffusion models. The optimally tuned model is further used for rendering high-quality skin lesion synthetic data with diverse and realistic characteristics, providing a valuable supplement and diversity to the existing training data. We investigate the impact of incorporating newly generated synthetic data into the training pipeline of state-of-art machine learning models, assessing its effectiveness in enhancing model performance and generalization to unseen real-world data. Our experimental results demonstrate the efficacy of the synthetic data generated through stable diffusion models helps in improving the robustness and adaptability of end-to-end CNN and vision transformer models on two different real-world skin lesion datasets.

Clinical Relevance—Synthetic medical imaging datasets play a crucial role in advancing healthcare by supporting education, research, algorithm development, and addressing privacy concerns. They contribute to the improvement of medical imaging technologies and the development of more accurate and robust diagnostic tools.

I. INTRODUCTION

Fatal skin disease, especially malignant skin cancer has emerged as a significant public health concern and is steadily gaining serious medical attention in recent years. The rising incidence and recent statistics of skin cancer [1] can be attributed to various factors, including increased ultraviolet (UV) radiation exposure due to changing environmental patterns and lifestyle habits. Sun exposure, indoor tanning, and inadequate protection against UV rays significantly contribute to the proliferation of skin cancer cases. Furthermore, an aging population, genetic predispositions, and a lack of awareness regarding skin cancer risk factors further exacerbate the issue. Skin cancer not only poses a substantial threat to individual health and well-being but also exerts a considerable economic burden on healthcare systems worldwide.

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Therefore, it is imperative to emphasize outreach efforts, promote sun-safe behaviours, and encourage regular skin examinations to mitigate the impact of this growing health problem. Early detection, timely intervention, and proactive preventive measures are essential in reducing the incidence and mortality rates associated with skin cancer. Lots of recent studies have shown shocking statistics [2], [3], [4] related to the increased number of skin cancer cases throughout the world.

Large scale medical imaging datasets play a vital role in development and deployment of efficient AI based computer aided diagnosis systems which can be effectively used by doctors and professionals for prompt diagnosis of various skin diseases [5] including malignant skin cancer [6]. Acquiring real-world skin lesion imaging data for detecting various skin disorders is indeed valuable for research and advancements in healthcare. However, acquiring such types of datasets is often challenging and further utilizing these datasets for research and analysis comes with several limitations as discussed in Section II that can impact the accuracy, diversity, and applicability in various contexts.

Keeping this in view, in this research work we have focused on utilizing AI-based generative stable diffusion models [7] for generating large-scale synthetic skin lesion data by unveiling the power of large text-to-image diffusion models [8]. This novel framework is referred to as Dermatoscopic Text-to-Image Model (Derm-T2IM) that uses natural language text prompts as an input and produces high quality malignant (including melanoma, basal cell carcinoma and squamous cell carcinoma) and benign (including nevus) lesion imaging data as the output. This framework can effectively comprehend the requirements of real-world large-scale skin imaging datasets. Incorporating image synthesis techniques into medical research can enhance data utilization, improve machine learning model performance, and accelerate advancements in the diagnosis, treatment, and understanding of various medical conditions. Further the data generated using diffusion based generative algorithms works as a reliable alternative to real world medical imaging data by providing various advantages such that synthetic images can be created to represent a wide range of variations in anatomy, pathology, dermoscopy, medical imaging modalities, and clinical scenarios. Secondly image synthesis allows for the creation of de-identified and privacy-preserving synthetic medical data, enabling secure sharing of data for research and collaboration without risking patient privacy or violating regulatory requirements. Further synthetic images can serve as a benchmark for evaluating the performance of various im-

age processing, reconstruction, and enhancement algorithms, enabling a fair comparison of different methodologies in a controlled setting. Last but not least generating synthetic data is often less expensive and time-consuming compared to collecting real patient data. This cost-effectiveness can accelerate research and development in the field of medical imaging.

In this work we have used a few shot data learning technique [9] via pretrained stable diffusion models [10] to address the challenges of learning patterns from a controlled amount of dermatoscopic data. The tuned model is then used for generating high-quality synthetic skin lesion data samples. The advantages of using few-shot adaptation of stable diffusion models include the ability to generate diverse and realistic data samples, which can lead to better generalization and performance on tasks with limited training data. To accomplish this task, we have employed DreamBooth [11] for tuning a customized text-to-image diffusion model capable of generating high quality skin lesion data. Section 3 details the proposed algorithm and complete working methodology proposed in this research work. The quality of generated synthetic data is cross-validated by fine-tuning two state-of-the-art deep learning classifiers which include vision transformers [12] and end-to-end pretrained CNN MobileNet-v2 [13] architecture. The complete experimental results are detailed in Section IV. The core contributions of this research work are listed below.

- Adaptation and optimal tuning of stable diffusion models with as low as 1400 images per class for developing a customized dermatoscopic text to image (T2I) diffusion models for rendering large scale synthetic dermatoscopic data based on user text prompts.
- The tuned T2I model is open-sourced via hugging face library [Link: <https://huggingface.co/Mali-Farooq/Derm-T2IM>] for research community and medical professionals to render more synthetic dermatoscopic skin lesion data as per their requirements.
- A novel large scale synthetic skin lesion data with various smart transformations which includes skin lesions of different classes, manipulating the size of moles, and rendering multiple moles, is open-sourced. The overall data comprises 3k malignant skin mole samples and 3k benign moles.
- Further validating the efficacy of newly generated skin lesion data by fine tuning vision transformers and end-to-end pre-trained CNN MobileNet V2 [13] with different settings. The performance of tuned classifiers is cross validated against ISIS Archive and HAM10000 skin cancer dataset with the highest overall accuracy of 93.03% using vision transformers.

II. BACKGROUND

Skin lesion detection and classification are crucial in dermatology for the early diagnosis of various skin diseases, including fatal skin cancers such as melanoma [14], [15], [16]. Deep Convolutional Neural Networks (CNNs), such

as EfficientNet, DenseNet, ResNet, and Inception, and MobileNet V2 have been widely used for robust skin lesion feature extraction and classification [17]. However deep learning models are often considered data-hungry because they typically require a large amount of training data to perform effectively. Table I provides a list of large-scale publicly available skin imaging datasets. However, real-world skin cancer datasets used in research and clinical applications can present several challenges and limitations. These problems can affect the quality, representativeness, and applicability of the data. Some common problems with large-scale real-world skin lesion include:

- **Data Quality and Variability:** Skin cancer datasets often suffer from varying image quality, lighting conditions, resolutions, and perspectives. Inconsistent image quality can affect the accuracy and reliability of machine learning models.
- **Data Imbalance:** Dermatoscopic datasets may exhibit imbalanced class distributions, with some types of skin lesions being more prevalent than others. This can bias machine learning models and affect their ability to accurately classify rare skin conditions.
- **Inter-Observer Variability:** Different medical professionals might interpret and label skin lesions differently. This can lead to discrepancies in the dataset, affecting the ground truth and potentially influencing the performance of machine learning algorithms.
- **Heterogeneity and Standardization:** Skin cancer datasets often come from diverse sources, resulting in heterogeneous data in terms of image formats, annotations, and metadata. Standardizing the data is challenging, but critical for effective analysis and model training.
- **Lack of Annotated Data:** Annotating skin lesion images requires expert dermatologists, and obtaining a large-annotated dataset can be time-consuming and expensive. Limited availability of annotated data hinders the training of accurate machine learning models.
- **Privacy and Ethical Concerns:** Medical datasets, including pigmented skin lesions, contain sensitive information about individuals. Ensuring privacy and complying with ethical guidelines while sharing or using such data poses a significant challenge. Further in the European Region (EU) General Data Protection Regulation (GDPR) sets stringent rules regarding the collection, processing, and storage of sensitive medical data.
- **Longitudinal Data:** Tracking the progression of skin lesions over time is essential for a comprehensive understanding of skin cancer. However, obtaining longitudinal data with consistent patient follow-ups can be difficult and may limit research insights.
- **Diversity of Skin Types and Ethnicities:** Skin cancer affects individuals with various skin types and ethnic backgrounds differently. The lack of diversity in the dataset may result in biased models that do not perform well for certain populations.
- **Limited Metadata and Clinical Information:** Many skin

TABLE I
DATASETS

S.No	Dataset Name	Dataset Details	Clinical Diagnostic Attributes
1	ISIC (International Skin Imaging Collaboration) Archive [18]	The ISIC Archive is one of the most comprehensive resources for dermatology-related image datasets. It includes a variety of skin lesion images, with labels for conditions like melanoma, nevus, and basal cell carcinoma.	nevus, melanoma, basal cell carcinoma, seborrheic keratosis, squamous cell carcinoma, actinic keratosis, pigmented benign keratosis, solar lentigo, dermatofibroma, vascular lesion, lichenoid keratosis, acrochordon, lentigo NOS, atypical melanocytic proliferation, AIMP, verruca, angioma, lentigo simplex, melanoma metastasis, other (33)
2	PH ² Dataset [19]	The PH ² Dataset consists of dermoscopic images of common pigmented skin lesions, along with ground truth data. It is commonly used for research in melanoma detection.	Common Nevus, atypical nevus, melanoma
3	HAM10000 (Human Against Machine with 10,000 training images) Dataset [20]	This dataset contains over 10,000 high-quality dermoscopic images with annotations for different skin lesion categories, including melanoma, and nevus. The overall dataset includes a representative collection of a wide range of important diagnostic categories in the realm of pigmented lesions	Clinic keratosis and intraepithelial carcinoma, Bowen's disease, basal cell carcinoma, benign keratosis-like lesions (solar lentigines / seborrheic keratoses and lichen-planus like keratosis, dermatofibroma, melanoma, melanocytic nevi and vascular lesions (angiomas, angiokeratomas, pyogenic granulomas and hemorrhage
4	Dermofit Image Library [21]	The dataset provides high quality skin lesion images for research purposes in computer science and medical imaging. The Dermofit Image Library is a collection of 1,300 focal high quality skin lesion images collected under standardised conditions with internal colour standards.	Clinic Keratosis, Basal Cell Carcinoma, Melanocytic Nevus (mole), Seborrheic Keratosis, Squamous Cell Carcinoma, Intraepithelial Carcinoma, Pyogenic Granuloma, Haemangioma Dermatofibroma, Malignant Melanoma
5	DermIS (Dermatology Information System) Dataset [22]	It is one of largest dermatology datasets available online. DermIS offers a collection of clinical images and cases related to dermatology.	Acrolentiginous Melanoma, Basal Cell Carcinoma, Hematoma, Melanocytic Nevus (mole), and others

cancer datasets lack detailed clinical information such as patient demographics, medical history, and other relevant factors. This limits the ability to conduct comprehensive analyses and develop more context-aware models.

- **Generalization to Real-World Settings:** Machine learning models, especially deep learning networks trained on controlled datasets may not always generalize well to real-world clinical settings. The models need to be robust and applicable to diverse clinical environments

for effective diagnosis and decision support.

Addressing these challenges requires collaboration among clinicians, data scientists, and researchers to improve data quality, standardization, and diversity in skin lesion datasets which is often very laborious and time-consuming process.

A. Data Augmentation in Medical Imaging using Computer Vision Algorithms

To address the challenges discussed in Section II, we have concentrated on working with digital image synthesis for dermatoscopy. It involves generating synthetic data, with precise

details of skin lesions. Further, dermatoscopic image synthesis serves various purposes, including augmenting datasets, simulating skin conditions, and enhancing the training of machine learning models for accurate skin lesion analysis. This can be attained by using various advanced image generation techniques, including generative adversarial networks (GANs) [23], variational autoencoders (VAEs) [24], physics-based simulations and recently emerged text-to-image stable diffusion models [25]. We can find several studies where the research community have particularly focused on using GANs for medical image synthesis [26], [27] however GANs are not inherently invertible. Unlike some generative models, such as VAEs, GANs do not provide a straightforward way to map generated data back to the latent space. This can be a limitation in critical applications such as medical data.

Synthetic image generation through a stable diffusion model can augment medical imaging datasets. This is especially valuable when there's a scarcity of data for specific conditions and when explicit diversity is required in existing medical imaging modalities. We can find published studies where the research community have proposed digital image synthesis for various medical imaging applications [28], [29], [30], [31]. Akrou et al. [28] proposed text to image synthesis method which is closest to our work which demonstrates the effectiveness of generative models, specifically probabilistic diffusion models (DPMs), in generating high-quality synthetic images of macroscopic skin diseases. This is achieved by conditioning the generation process on text prompt inputs, allowing for fine-grained control of the image generation process. However, there approach depends on utilizing large number of data samples for robust fine-tuning of pretrained stable diffusion models. Further authors do not discuss much about the computational efficiency of the proposed model. Moreover, at this stage the authors have not open-sourced the synthetic macroscopic skin diseases dataset and tuned models. In another study [32], authors demonstrated that pre-trained Stable Diffusion models, originally trained on natural images, can be adapted for various medical imaging modalities by training text embeddings with textual inversion. The authors have validated the approach on three different medical imaging modalities which include prostate MRI images, chest x-ray and lymph node sections histopathology images. Further experimental analysis shows that synthetic data helps by achieving a 2% increase in diagnostic accuracy (AUC) for detecting prostate cancer on MRI, from 0.78 to 0.80. Shavlokhova et al. [33] fine-tuned the GLIDE stable diffusion model on different clinical diagnostic entities, including melanoma and benign melanocytic nevi. Authors used 10,015 dermoscopic images for this purpose thus relying on sufficient seed training data. They created photorealistic synthetic samples of each diagnostic entity using tuned models. For evaluation purposes authors have used various image quality metrics which includes SSIM, PSNR, and FID scores.

Our work extends on this by exploring few-shot medical image generation, by training with limited yet high-quality

data samples. Additionally, we demonstrate the efficacy of the proposed approach by indulging various smart transformations using text prompts thus reflecting the effectiveness of tuned text-to-image diffusion model. In contrast to most other studies, we intentionally do not train from scratch and use small datasets to explore the feasibility of fine-tuning diffusion in low-data and low-compute environments.

III. BUILDING DERM-T2IM

This section will describe the proposed methodology adapted in this work for generating large scale high quality synthetic skin lesion data. As mentioned in Section I we have primarily focused on using small amount of skin lesion representation and further few shot [11] learning method for fine-tuning text-to-image stable diffusion models. Probabilistic Diffusion models [34] are a class of AI based generative models used to model complex probability distributions. They capture the dynamics of data by iteratively diffusing a data point such as an image towards the data distribution through a series of transformations. These models are particularly useful for generating realistic and high-quality data samples. Stable diffusion models based on a particular type of diffusion model called Latent Diffusion model [35], are a variant of diffusion models designed to provide more stability and improved training characteristics. They aim to address issues like training instability and mode collapse often encountered with standard diffusion models. Fig. 1 shows the block diagram representation of the proposed methodology.

As it can be seen from Fig. 1 the first step is preparing the input data (seed data) for training the models. In this work we have used International Skin Imaging Collaboration (ISIC) Archive data [18]. The ISIC Archive was created to establish an extensive public repository of meticulously annotated high-quality skin images. This resource provides clinicians and educators in enhancing diagnostic capabilities and delivering clinical assistance for the detection of skin cancer. The selected set data is divided in binary classes which includes malignant and benign mole data. A total of 2800 data samples with 1400 benign skin mole images and 1400 malignant skin moles were used for this purpose. The second step includes data preprocessing thus preparing the data for feeding it to the data loader. Data preprocessing incorporates resizing the image size to 512×512 as input tensor shape and further removing data clutters such that unwanted skin hairs covering the lesion region of interest thus properly focusing on lesion area. To accomplish this, we have used open-source dull razor software [36]. Fig. 2 shows the hair removal processing results on two sample cases using dull razor application.

The third and fourth step include using the processed data and passing it to the data loader. The processed data is divided in two different data loading concepts which includes malignant skin lesion data and benign skin lesion data.

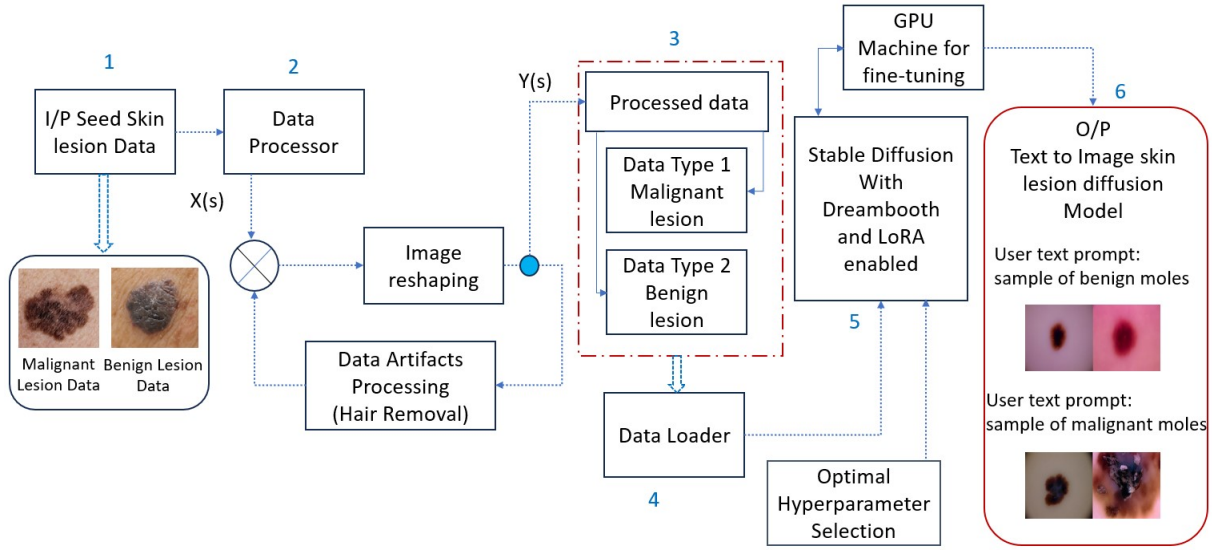


Fig. 1. Comprehensive block diagram representation of proposed methodology.

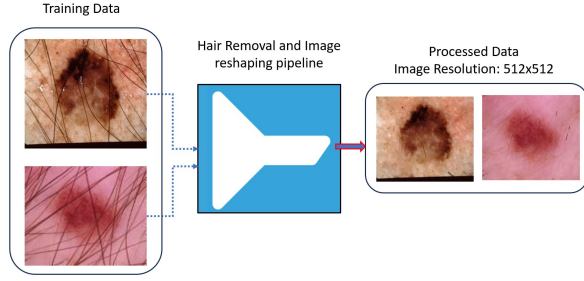


Fig. 2. Skin mole hair removal data processing outputs on two different cases generated using dull razor software.

A. Stable Diffusion Training Components

As mentioned in Section I we have incorporated Dreambooth tool which is based on few shot data learning methods for fine-tuning text-to-image stable diffusion models (T2IM). The core idea is to condition the image generation process on a given image text description. Equation 1 represents the simplified conceptual equation to illustrate the T2IM. Further Low-Rank Adaptation (LoRA) is enabled for performing stable training process. The details of these tools are provided below.

$$I = G(T, \theta) + \varepsilon \quad (1)$$

Where G is the generator function that takes the textual description T and model parameters θ to produce an image I . ε represents random sampling noise added during the image generation process.

Dreambooth

Dreambooth, which is being developed by Google research and Boston University researchers offers the unique advantage of utilizing a small set of seed training data depicting a particular data class/subject, by adapting a pre-trained text-to-image Imagen [8] model to associate a distinct identifier with that specific subject. After integrating the subject into

the model's output space, this identifier becomes a tool for generating new images of the subject within various contextual settings. The approach leverages the inherent semantic knowledge within the model, along with a novel self-generated class-specific prior preservation loss, thus allowing to create representations of the subject in a wide range of scenes, poses, perspectives, and lighting conditions that were not present in the original reference images. In this work we have adapted the same tool for generating photo-realistic skin lesion data thus validating its effectiveness in medical imaging applications.

LoRA

Utilizing Low-Rank Adaptation [37] in stable diffusion during the training phase is a valuable tool for improving the performance and stability of generative models. Equation 2 shows the incorporating LoRA while fine-tuning stable diffusion models. β is the merging ratio. It works by scaling beta from 0 to 1. Setting beta to 0 is the same as using the original model and setting beta to 1 is the same as using the fully fine-tuned model. It allows the efficient incorporation of new data while leveraging the pretrained knowledge in the model's parameters. LoRA enhances training efficiency and significantly reduces the computational hardware requirements, up to threefold, particularly when employing adaptive optimizers. This is achieved by eliminating the need to compute gradients and manage optimizer states for the majority of parameters. As an alternative the focus is on optimizing the compact, injected low-rank matrices, resulting in streamlined and resource-efficient training process. This approach is especially significant in scenarios where retraining from scratch is impractical and resource intensive. In this work we have used Low-rank adaptation for optimal fine-tuning of pretrained stable diffusion models with a lower-dimensional representation based on new data type, thus without fully retraining the entire model.

Algorithm 1 Derm-T2IM Training Algorithm

Input: Text descriptions T ; Few-shot image data $x(t)$ (Binary Classes: 0, 1).

- 1: Preprocess the text descriptions and few-shot image data $x(t)$;
- 2: Initialize the model parameters (θ) ;
- 3: **for** $i = 1$ to n **do**
- 4: Encode the text descriptions T using a text encoder ι ;
- 5: Conditioning vector $C = \iota(P)$ generated using a text encoder ι and a text prompt P ;
- 6: Encode the few-shot image data using an image encoder ε ;
- 7: Combine the encoded text ι and image features;
- 8: Define the noise Scheduler δ_t ;
- 9: Generate initial images using a generator network $G(T, \theta)$;
- 10: Compute the loss between the generated images and the real images;
- 11: Update the generator network $G(T, \theta)$ using backpropagation and gradient descent.
- 12: **end for**

Output: $I = G(T, \theta) + \varepsilon$, Generated images from new text descriptions T and random sampling noise.

$$W(x) = W_0 + \beta \Delta W \quad (2)$$

Tuning Derm-T2IM

Our primary focus is building customized diffusion model using a pre-trained text-to-image diffusion model, which takes a few-shot image data $x(t)$, along with a conditioning vector C generated through a text encoder based on a given text prompt P . This model, when provided with these inputs, and further trained using the mentioned training components, produces fully synthetic images. The below pseudocode shows the complete training description for building Derm-T2IM.

B. Training Parameter Selection

The next step includes selecting an optimal set of training parameters. Initially we conducted a lot of experiments to analyse the effect of different settings in Dreambooth since it plays a critical role in achieving successful and effective training results. Although the training process relies on numerous training parameters, in this section, we have discussed the most important parameters which are detailed in Table I that were shortlisted during the fine-tuning phase of stable diffusion using Dreambooth tool.

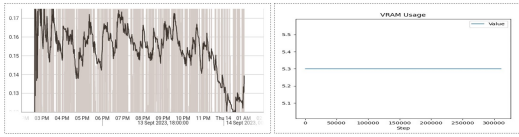


Fig. 3. Loss graph of tuned Derm-T2IM with final loss value of 0.1394 and overall training time of 10.4 hours. The second graph shows the average amount of video RAM i.e., 5.3 GB required during the whole training process.

As it can be observed from Table I that we have selected Discrete Denoising Scheduler (DDS) as the noise scheduler. The noise schedule is a critical component of diffusion models as it simulates the gradual diffusion process from noisy to clean data. It is important to select an effective noise scheduler to ensure that the model converges and generates

TABLE II
HYPER-PARAMETER SELECTION

Training Parameter	Details
Training Batch Size	2
Training Epochs	120
Gradient accumulation steps	1
Optimizer	8-bit, AdamW
Minimum learning rate	$1e^{-6}$
Learning rate	$2e^{-6}$
LoRA learning rate	$1e^{-4}$
Noise Scheduler	Discrete Denoising Scheduler(DDS)
Mixed precision selection	FP16
Learning Scheduler	Constant with warmup
Model input size (Resolution)	512

realistic data samples. The Discrete Denoising Scheduler (DDS) is a scheduling strategy used in denoising diffusion probabilistic models (DDPMs) [34] to determine how the noise level should evolve during the denoising process. DDS provides a discrete set of noise levels at each step and helps control the level of noise introduced during the training process. The DDS representation as shown in equation 3 is a discrete formulation used to schedule the noise levels at each training step. This equation defines how the observation at the current time step (x_t) is obtained from the previous observation (x_{t-1}) by adding noise ($\sigma_t * z_t$). The noise level σ_t is a key component of the DDS and is typically scheduled to decrease over time during training.

$$x_t = x_{t-1} + (\sigma_t * z_t) \quad (3)$$

Where x_t is the noisy observation at time step t . x_{t-1} is the observation at the previous time step ($t - 1$). σ_t is the noise level or standard deviation at time step t . z_t is a sample from the standard normal distribution (mean 0, standard deviation 1). It is designed to be computationally efficient while achieving high-quality sample generation.

As the optimizer, we have selected ‘AdamW’ [38] with ‘Constant with Warmup’ as the learning scheduler. ‘Constant with Warmup’ is a commonly used learning rate scheduling technique in various deep learning applications. It helps strike a balance between stability and speed in training/tuning deep learning models. Finally, FP16 model precision is selected. FP16 is a valuable tool in the optimization of deep learning and high-performance computing workloads. When used judiciously and with consideration of its limitations, it can significantly improve the efficiency and speed of computations, making it particularly relevant for large-scale machine learning based image rendering operations and scientific simulations.

C. Image Inference Samplers

Once the model is trained using few shot data learning and optimal selection of training parameters, the sixth step as shown in Fig. 1 includes rendering new synthetic dermatoscopic data using various image sampling methods. In this research study we have used three different text-to-image sampling methods which incorporate Euler, Euler a and PLMS methods [39]. Euler sampling stands as the simplest possible sampling method, sharing mathematical equivalence with Euler’s method for solving ordinary differential equations. It operates entirely deterministically, devoid of any introduction of random noise during the sampling process. Euler ancestral (Euler a) sampler is like Euler’s sampler. But at each step, it subtracts more noise than it should and adds some random noise back to match the noise schedule. The denoised image depends on the specific noise added in the previous steps. So, it is an ancestral sampler, in the sense that the path the image denoises depends on the specific random noises added in each step. Further we have used a more recent sampling technique known as Pseudo Numerical Methods for Diffusion Models (PLMS) on Manifolds are computational techniques designed to approximate and simulate diffusion processes on mathematical manifolds. These methods are particularly useful for modeling complex data distributions and conducting probabilistic modeling and generative tasks on manifolds. They allow for the estimation of data distribution dynamics and the generation of realistic data samples by discretizing the diffusion process into numerical steps. The term ‘pseudo’ indicates that these methods approximate the behaviour of diffusion processes, typically using numerical integration or sampling techniques.

IV. EXPERIMENTAL RESULTS

All our experiments were conducted by integrating dream-booth extension with Stable Diffusion Web User Interface which is based on the Gradio python library for developing customized dermatoscopic text-to-image diffusion models. The complete environment was set up on a workstation machine equipped with the RTX3090Ti graphics card with 24GB of dedicated graphics memory.

A. Fine Tuning Results

Initially, we used various training parameters as the ablation study to analyze the model tuning results. Finally,

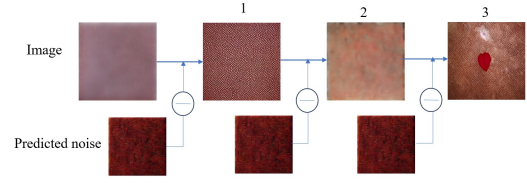


Fig. 4. Learning samples with predicted random noise while performing the fine-tuning process of Derm-T2IM: First Sample extracted at 12990 steps and last sample extracted at 311760 steps.

we shortlisted the set of training parameters as mentioned in Table II for performing optimal convergence and model generalization on our custom skin lesion data. The model was fine-tuned with a batch size of 2 for 120 epochs. We used a high learning rate of $2e-6$ and a low learning rate of $1e-6$ and further no prior preservation was used. It is important to discuss that we have used 8-bit Adam, along with fp16 gradient accumulation to reduce memory requirements and run these experiments on GPUs with 24 GB of memory. The overall training process was completed in 10.4 hours. Fig. 3 shows the loss graph of the tuned model extracted via the Tensorboard visualization tool and the average VRAM required during the fine-tuning process. Whereas Fig. 4 shows the learning samples during the model fine-tuning process at various steps. Since in this research work, we are more interested in employing a pre-trained text-to-image diffusion model for validating its effectiveness in medical imaging application we have used frozen text encoder resulting in small, personalized token embeddings.

B. Derm-T2IM Rendering Results using different sampling methods

The fine-tuned Derm-T2IM model is subsequently employed to generate novel synthetic skin lesion data. This is achieved by using three different image sampling methods as discussed in Section III-C. Fig. 5 shows the results along with user text prompts.

As it can be observed from Fig. 5 that quality of rendered data improves by increasing by sampling steps. Therefore, we have mostly used sampling steps between the scale of 22-26 for generating large scale skin lesion data. The overall dataset was generated with image resolution of 512×512 and stored in PNG format. PNG is lossless compression and thus retains all the data in the file during the compression process, which is important when we have to resize the data for further use for downstream machine learning tasks. Furthermore, in the image rendering process, the classifier-free guidance scale (CFG) scale was configured to a value of 7. Typically, this setting determines the degree to which the generated image aligns with the user’s input prompt. Higher values tend to maintain the image’s proximity to the user’s prompt.

C. Smart Image Transformation using Diversified Text Embeddings

To further validate the potential of optimally tuned Derm-T2IM architecture we have produced pigmented lesion data

Image Sampler: Euler, Sampling Steps: 20, Text Input: Sample of benign moles

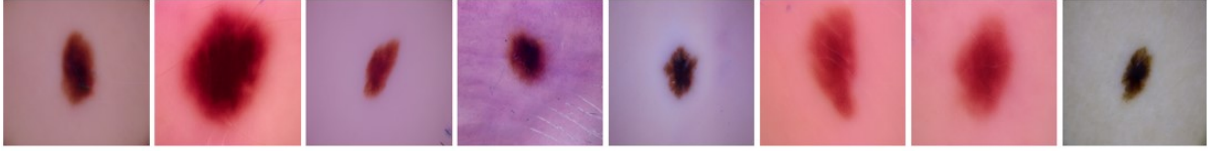


Image Sampler: Euler, Sampling Steps: 20, Text Input: Sample of malignant lesion

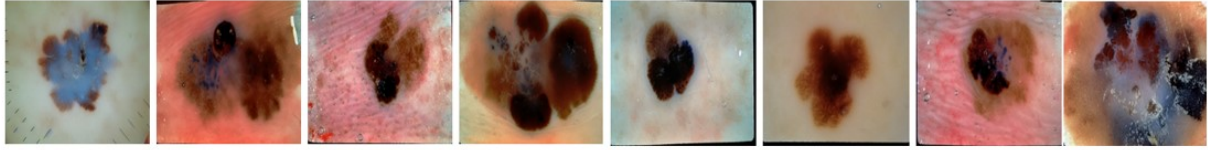


Image Sampler: Euler a, Sampling Steps: 24, Text Input: Generate benign skin moles

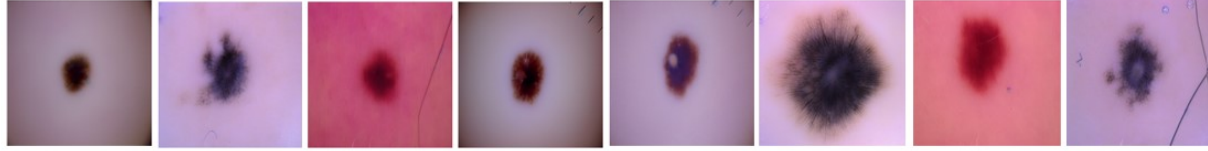


Image Sampler: Euler a, Sampling Steps: 22, Text Input: Generate malignant skin data samples

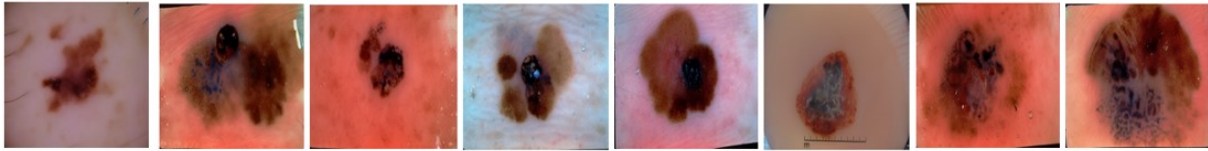


Image Sampler: PLMS, Sampling Steps: 22, Text Input: Benign lesion mole on human skin

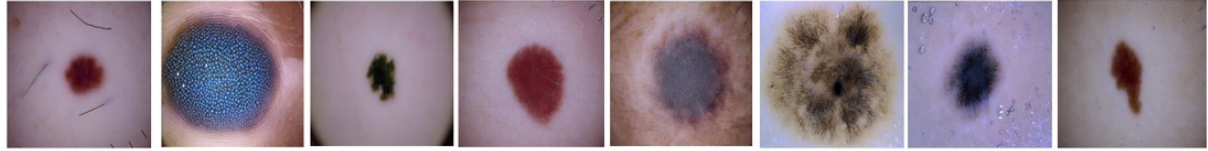


Image Sampler: PLMS, Sampling Steps: 24, Text Input: Render malignant skin data samples

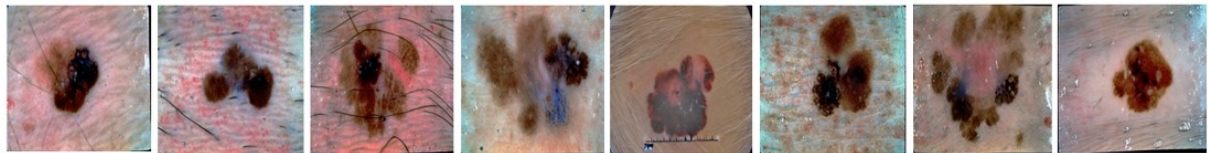


Fig. 5. Rendered outputs showing newly generated benign and malignant skin mole data using Derm-T2IM. The first, third and fifth rows show the results of benign data generated via Euler, Euler a, and PLMS sampling method whereas second, fourth and six rows show the malignant lesion inference results generated via Euler, Euler a, and PLMS samplers.

with advanced transformation based on user text inputs/prompts. These smart data transformations are produced by employing three different sampling methods as discussed in Section III-C and varying sampling steps. The purpose is to generate data that can help the research community and medical professionals to better understand and classify the moles under peculiar circumstances. These transformations include manipulating the size of moles, rendering multiple moles, generating dermatoscopic data embedded on different skin colour. The detailed results are demonstrated in Fig. 6.

It can be observed from Fig. 6 that a tuned model can be efficiently used for producing a variety of data without undergoing any further specific training/ tuning process. For now, we have tested the Derm-T2IM with these types of

advanced augmentations however further diversity can also be rendered depending on specific use cases.

D. Novel Large Scale Skin Lesion Data and Open-Sourced Derm-T2IM Stable Diffusion Architecture

Through this work we have open-sourced a large scale synthetic dermatoscopic data by employing the fine-tuned stable diffusion model (Derm-T2IM). This is achieved by storing the trained model using a simpler yet faster safetensors format. In the second phase the Derm-T2IM safetensor file is uploaded via text to image inference engine for rendering novel synthetic skin lesion dataset. The overall dataset consists of binary classes i.e., ‘benign’ and ‘malignant’ moles which is generated by providing the different user prompts

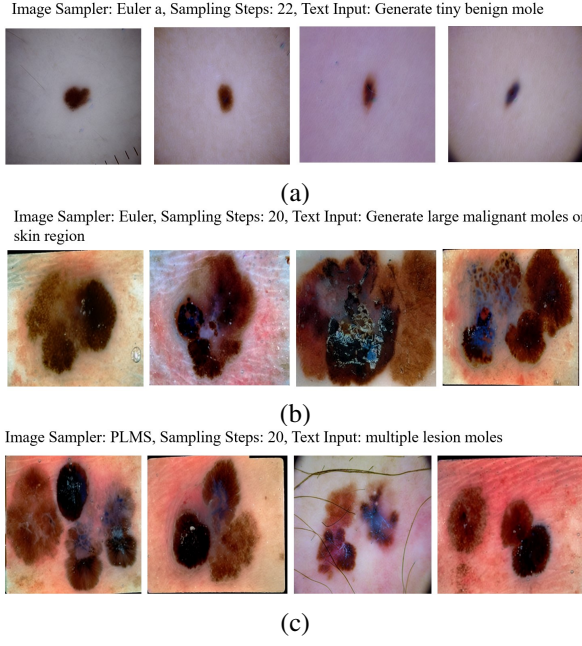


Fig. 6. Derm-T2IM smart transformation outputs: a) shows the tiny benign mole results using Euler a method, b) shows the sample images of large malignant mole, c) shows the transformation results of multiple lesion moles.

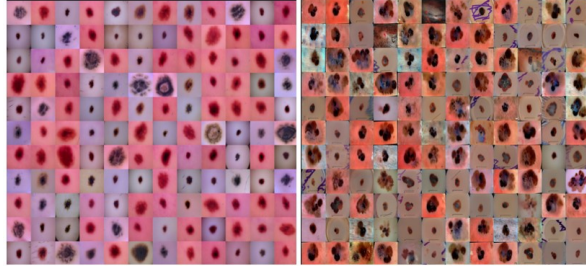


Fig. 7. Large scale novel synthetic skin lesion dataset generated using Derm-T2IM Stable Diffusion, a) 11 $\times$ 12 image grid showing the sample of benign lesion data generated using euler, euler a and PLMS sampling methods, b) show the sample of malignant lesion data generated using Euler, Euler a and PLMS sampling methods.

TABLE III
DATASETS DETAILS

Data Class	No of Samples	Sampling Methods	Sampling steps	Image Resolution and Image compression
Benign	3k	Euler, Euler a, PLMS	20-26	512 $\times$ 512, PNG
Malignant	3k			

as shown in Fig. 7. Further Fig. 7 shows the image grid of data samples collected from novel synthetic skin lesion data whereas Table III shows the complete dataset attributes.

It's imperative to note that during the generation and storage of a large-scale synthetic skin lesion dataset, we carefully excluded a set of images containing noisy artifacts as demonstrated in Figure 8. This process resulted in a pristine and high-quality dataset. The complete Synthetic Derm-T2IM Dataset is available via our GitHub repository:



Fig. 8. Samples of discarded data with noisy artifacts.

V. SYNTHETIC DATA VALIDATION

In the second phase of this research study, we aim to assess the quality and utility of the data generated via Derm-T2IM. We do this by training and validating two state-of-the-art skin lesion classifiers, which incorporate Vision Transformer and MobileNet V2 CNN. Vision Transformers (ViTs) have proven to be highly effective for medical image classification tasks. Their ability to capture complex patterns and features within medical images has made them a valuable tool in healthcare and medical research application. In this study we have used the 'google/vit-base-patch16-224-in21k' model developed by Google. It's a base-sized model that takes 224 $\times$ 224 pixel images divided into 16 $\times$ 16 pixel patches as input and was pre-trained on a dataset with 21,000 classes [12]. Similarly, MobileNetV2 is a convolutional neural network architecture designed for efficient on-device image classification and computer vision tasks. It introduces the concept of inverted residuals, which involves the use of lightweight depth-wise separable convolutions with linear bottlenecks. This design enhances the efficiency of the network. Further the linear bottlenecks help to maintain low-dimensional embeddings between layers, allowing for better information flow and reducing the risk of information loss. Such pre-trained models can be fine-tuned via transfer learning for various medical imaging applications and as robust feature extractors for image analysis. Fig. 9 shows the adapted training and validation methodology for both the skin lesion classifiers.

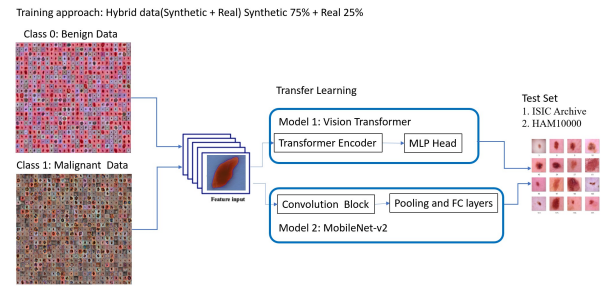


Fig. 9. Derm-T2IM synthetic data validation: adapted hybrid training data approach for better fine-tuning of MobileNet-V2 and Vision Transformers Models.

It can be observed from Fig. 9 that we have used hybrid data training approach for the purpose of transfer learning and further cross validating the tuned networks on a set of testing data collected from real-world skin cancer datasets. Transfer learning is a powerful technique in deep learning where pre-trained models, initially trained on large-scale datasets, are fine-tuned for new tasks/ domains [40] without the requirement of training the complete network

TABLE IV
TEST ACCURACY

		Dataset-1 ISIC Archive	Dataset-2 HAM10000
Baseline Model Accuracy %	MobileNet	78.89	70.33
	ViT	90.74	81.50
Hybrid Model Accuracy %	MobileNet	81.35	73.58
	ViT	93.03	85.02

from scratch. When it comes to computer vision, particularly image classification tasks, transfer learning can be applied to both Vision Transformers (ViTs) and Convolutional Neural Networks (CNNs). Hybrid training data comprises a larger division ($\approx 75\%$) of synthetic data with small of portion of real data samples ($\approx 25\%$) to incorporate data diversity, avoid model over fitting, and enlarge the overall training data size. The performance of tuned binary classifier models trained on hybrid data is compared with baseline models which are fine-tuned using only real dermoscopic data samples. For comparison analysis we have perform the cross-validation test on a set of unseen images collected from two different datasets which includes ISIC archive¹ and HAM10000 [41] dataset. The below graphs show the accuracy comparison analysis on validation set of training data of both the tuned networks.

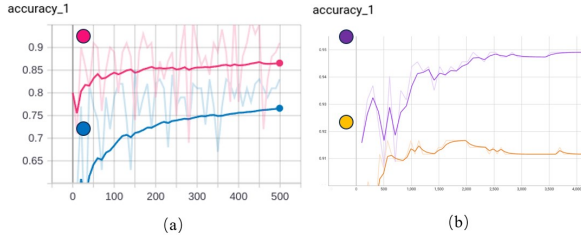


Fig. 10. Accuracy comparison computed on the validation set of training data. a) MobileNet: The pink graph shows the model tuned on hybrid data with validation accuracy of 86.57% whereas the blue graph shows the baseline model trained on only real-world skin data with validation accuracy of 76.61%. b) ViT: The purple graph shows the model tuned on hybrid data with validation accuracy of 94.74% whereas the orange graph shows the baseline model trained on only real-world skin data with validation accuracy of 91.16%

It can be observed from Fig. 10 that hybrid data approach which comprises of blended mixture of real and synthetic data samples helps in robust tuning of both CNN and vision transformer models. Further we have used a unseen test dataset acquired from two different real-world skin cancer datasets to perform the rigorous cross validation of both the networks as shown in Table IV The results demonstrates that models trained on hybrid data outperforms the baseline models by achieving better testing accuracy with nearly an improvement of 2.46% in case of mobilenet and 2.29% in case of ViT model.

In addition, to this we have performed two other validation tests. The first test includes precise mole segmentation without the need for additional training. For this purpose, we have

incorporated zero shot Segment Anything model developed by Meta AI² to perform mole segmentation on Derm2-T2IM dataset samples. The test was carried out on more than 100 different cases from both the classes. The segmentation results demonstrated in Fig. 11 shows the robustness of pretrained model on novel systematic synthetic datasets. In the second phase, we have performed detection as well as classification test on Derm-T2IM dataset by incorporating pretrained SSD YOLO-V8³ model trained on real world ISIC archive dataset⁴. In this case the cross-validation test was performed on 200 random samples selected from Derm-T2IM dataset. The model achieves the overall test accuracy of 72%. The inferred result on four different cases shortlisted from synthetic dataset is depicted in Fig. 12.

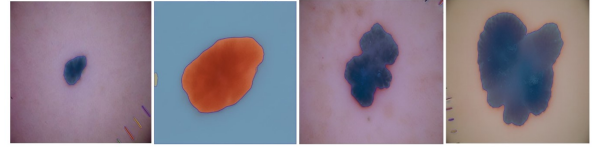


Fig. 11. Mole segmentation results of four different cases using zero shot Segment Anything model. The first two images show precise mole boundary and region of interest segmentation on benign data whereas the third and fourth shows the segmentation results on malignant data.

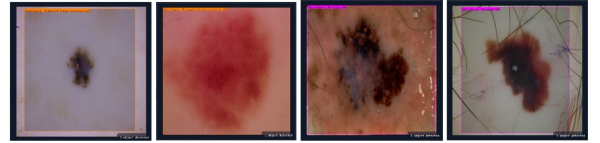


Fig. 12. Mole detection and classification results using SDD Yolo-v8 tuned on ISIC archive datasets. The first two images are detected and classified as benign keratosis with 75% and 86% confidence scores whereas the third and four cases are classified as melanoma with confidence scores of 60% and 63%.

VI. CONCLUSION AND FUTURE WORK

In this work, we have proposed an efficient text-to-image stable diffusion model based on few-shot learning with a small number of dermoscopic medical data representations for rendering high-quality synthetic skin lesion data. The generated data showcases the superior generative abilities of probabilistic diffusion models in creating diversified skin lesion images of different categories. Further, the efficacy of the generated data is validated through the integration of a hybrid training data approach, involving a blend of small amount of real and synthetic data. This approach enhances the robustness of training Convolutional Neural Network (CNN) and Vision Transformer models using the majority of synthetic data blended with real data. The results demonstrate that utilizing hybrid data contributes to achieving more precise validation and test outcomes on unseen real test data sourced from various publicly available skin cancer

²<https://segment-anything.com/demo>

³<https://github.com/ultralytics/ultralytics>

⁴<https://universe.roboflow.com/ai-eds7n/skin-cancer-recogniser>

¹<https://www.isic-archive.com/>

datasets. As conclusion the synthetic dermoscopic images are a valuable resource for dermatologists, researchers, and machine learning practitioners. They contribute to the development of more accurate diagnostic tools, better training for healthcare professionals, and the improvement of skin lesion analysis algorithms, ultimately enhancing the early detection and management of skin conditions.

As the possible future research directions, the existing system can be retrained to expand its generating capabilities thus rendering more heterogeneous data such as specific type of malignant neoplasms, other type of skin diseases such that acne cystica, bullous pemphigoid, and generating data of various degree of burn/ scald cases which involve the entire epidermis, resulting in erythema and blistering. Moreover, we can explore to develop methods to allow users more control over the generation process, enabling interactive and user-guided content synthesis.

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